THE UNITED REPUBLIC OF TANZANIA
THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT
ANNUAL JOINT EXAMINATION

FORM THREE
032 CHEMISTRY
MARKING SCHEME NOVEMBER 2024

SECTION A (16 marks)

1. Multiple choice (10marks @ 01mark)

QNS	i	ii	iii	iv	v	vi	vii	viii	ix	x
ANS	D	A	D	C	B	C	A	A	A	E

2. Match the items (06marks @ 01mark)

LIST A	i	ii	iii	iv	v	vi
LIST B	C	G	A	B	F	D

SECTION B (54 marks)

3. (a) (i) Enable a laboratory user to know how to use and handle the chemical at the time of doing experiment. *01½ Mark*

(ii) It helps to prevent accidents in the laboratory *01½ Mark*

(b)

(i) At room temperature, covalent compounds are in a liquid or gaseous state While electrovalent compounds are in crystalline form. *01½ Mark*

(ii) The melting and boiling points of covalent compounds are usually low While the melting and boiling points of electrovalent compounds are high. *01½ Mark*

(iii) Covalent compounds are insoluble in water While electrovalent compounds are soluble in water.

(iv) Covalent compounds do not conduct electricity While electrovalent compounds conduct electricity in molten state *01½ Mark* *01½ Mark*

4. (a) (i) The mass of an element liberated by 1 coulomb of electricity during electrolysis is called electrochemical equivalent of that element while the mass of an element deposited or liberated by 1 faraday during electrolysis is called chemical equivalent of that element. (1½ marks)

(ii) When the same quantity of electricity is passed through solutions of different electrolytes, the relative numbers of moles of the elements deposited are inversely proportional to the charges on the ions of each of the elements respectively. (01½ mark)

(b) Data given

Copper

$m_{Cu} = (Ar \times It)/(V \times F)$ (01 mark)

$It = (m_{Cu} \times F \times V)/Ar$ (01 mark)

$Q = (6.4 \times 96500 \times 2)/64$ (01 mark)

(0½ mark)

=19,300 C

Silver
m_s = (Ar x It)/(V x F) (01 mark)
= (108 x 19300)/96500 (01 mark)
= 21.6g (0½ mark)

5. (a)
Solution: One amendment is that, the atom, which was considered to be indivisible during Dalton's time, was discovered to be composed of still smaller sub-atomic particles. (02 marks)
Another amendment is that atoms of the same element may have different masses. This is due to the presence of isotopes. Isotopes are atoms of the same element which have different mass numbers.
(02 marks)

(b)
Solution
Percent abundance of $^{10}B \times {}^{10}B$ + Percent abundance of $^{11}B \times {}^{11}B$ = periodic table atomic mass
(02 marks)

$\frac{20 \times 10.01}{100} + \frac{80 \times 11.01}{100} = 10.81$ (01 mark)
(02 marks)

6. (i) 2 moles of $Ca(NO_3)_2 \equiv$ 4 moles of $NO_{2(s)}$
1 mole of $Ca(NO_3)_2 \equiv$? *00½ Mark*

$\frac{1\ mole\ of\ Ca(NO_3)_2}{2\ moles\ of\ Ca(NO_3)_2} \times$ 4 moles of $NO_{2(g)}$ *00½ Mark*

= 2 moles of $NO_{2(g)}$
∴ It requires 2 moles of $NO_{2(g)}$ *00½ Mark*

(ii) No. of moles of $Ca(NO_3)_2 = \frac{mass\ of\ Ca(NO_3)_2}{mass\ of\ Ca(NO_3)_2}$ *00½ Mark*

$= \frac{65.6g}{(40+(14\times2)+(48\times2)}$ *00½ Mark*

$= \frac{65.6g}{164g/mol}$ *00½ Mark*
= 0.4 moles of $Ca(NO_3)_2$
Then:
2 moles of $Ca(NO_3)_2 \equiv$ 1 mole of $O_{2(g)}$
0.4 moles of $Ca(NO_3)_2 \equiv$? *00½ Mark*

$= \frac{0.4}{2} \times 1$ mole of $O_{2(g)}$
0.2 mole of $O_{2(g)}$ *00½ Mark*
Finally:
Volume of $O_{2(g)}$ = No. of moles of $O_{2(g)}$ × Q.M.V
= 0.2 × 22.4 dm^3/mol
= 4.48 dm^3
∴ Volume of $O_{2(g)}$ =4.48 dm^3 *00½ Mark*

(iii)

2 moles of $Ca(NO_3)_2 \equiv$ 2 moles of $CaO_{(s)}$

0.4 moles of $Ca(NO_3)_2 \equiv$? *00½ Mark*

$= \frac{0.4}{2} \times 2$ moles of $CaO_{(s)}$

0.4 moles of $CaO_{(s)}$ *00½ Mark*

Then

Mass of $CaO_{(s)}$ = No. of moles of $CaO_{(s)}$ × molar mass

= (0.4 mol × 56) g/mol

= 22.4 g *00½ Mark*

∴ <u>mass of CaO = 22.4 g</u>

(iv)

2 moles of $Ca(NO_3)_2 \equiv$ 5 moles of gaseous products

? ≡ 5 moles of gaseous products

$= \frac{5}{5} \times 2$ moles of $Ca(NO_3)_2$ *00½ Mark*

2 moles of $Ca(NO_3)_2$

Finally

Mass of $Ca(NO_3)_2$ = No. of moles of × molar mass

= 2 mole × 164 g/mol

= 328 g *00½ Mark*

∴ <u>Mass of $Ca(NO_3)_2$ = 328 g</u>

(v) No. of moles of $NO_{2(g)} = \frac{Vol.of\ O_{2(g)}}{G.M.V}$ *00½ Mark*

No. of moles of $NO_{2(g)} = \frac{44.8\ dm^3}{22.4\ dm^3/mol}$

= 2 moles of $NO_{2(g)}$ *00½ Mark*

Then:

2 moles of $Ca(NO_3)_2 \equiv$ 4 moles of $NO_{2(g)}$

? ≡ 2 moles of $NO_{2(g)}$

$\frac{2}{4} \times 2$ moles of $Ca(NO_3)_2$

= 1 mole of $Ca(NO_3)_2$ *00½ Mark*

Finally

Mass of $Ca(NO_3)_2$ = No. of moles of × molar mass

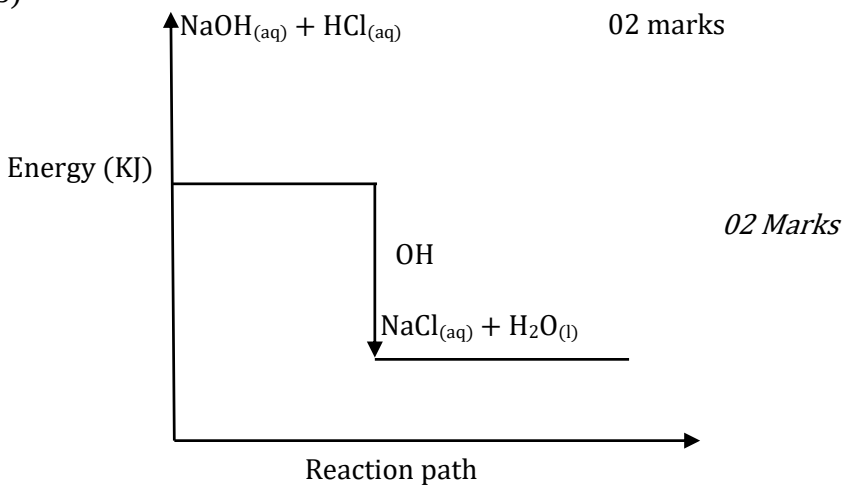
= 1 mole × 164 g/mol

= 164 g *00½ Mark*

∴ <u>Mass of $Ca(NO_3)_2$ = 164 g</u>

7. (a) the type of the reaction taking place is exothermic

(b)



02 marks

02 Marks

- Treating insects stings and bites
- Relieving indigestion
- Soil treatment
- Treating factory wastes
- Manufacturing fertilizers
- Neutralizing accidental spills
- Preventing formation of acid rains

Any four @01½ = 06Marks

8. (a) Titrant is a solution with a known concentration; during titration a titrant is placed in burette While Analyte is a solution with unknown concentration; during titration the analyte is placed in a conical flask. *03 Marks*

(b) $M(OH)_{2(aq)} + 2HCl_{(aq)} \rightarrow MCl_{2(aq)} + 2H_2O_{(l)}$

Given: mass of $M(OH)_2$ = 5.8g

Molarity of HCl = 0.2M

Vol. of soln = 1 dm^3

Number of moles of HCl *00½ Mark*

From:

Molarity = $\frac{\text{Number of moles}}{\text{Volume of solution}}$ *00½ Mark*

No. of moles = 0.2 mol/dm^3 × 1 dm^3

= 0.2 mol *00½ Mark*

Find number of moles of $M(OH)_2$

1 mole of $M(OH)_2 \equiv$ 2 moles of HCl

? ≡ 0.2 moles of HCl

$x = \frac{0.2 \text{ of moles HCl} \times 1 \text{ mole of } M(OH)_2}{2 \text{ moles of HCl}}$ *00½ Mark*

= 0.1 mole of $M(OH)_2$ *00½ Mark*

Find Mr of $M(OH)_2$

Molar mass = $\frac{\text{Mass of } M(OH)_2}{\text{No. moles of } M(OH)_2}$ *00½ Mark*

(i) Molar mass $M(OH)_2 = \frac{5.8g}{0.1 \text{ mol}}$ *00½ Mark*

58 g/mol of $M(OH)_2$

∴Molar mass of $M(OH)_2$ = 58g/mol *00½ Mark*

(ii) Find atomic mass of $M(OH)_2$
From
Mr of $M(OH)_2$ = ∑R.A.M *00½ Mark*
58g/mol = [(M) + (16 + 1) × 2] g/mol *00½ Mark*
58 = 34 + M
58 – 34 = M
M = 24 g/mol
∴ the atomic mass of M is 24 *00½ Mark*

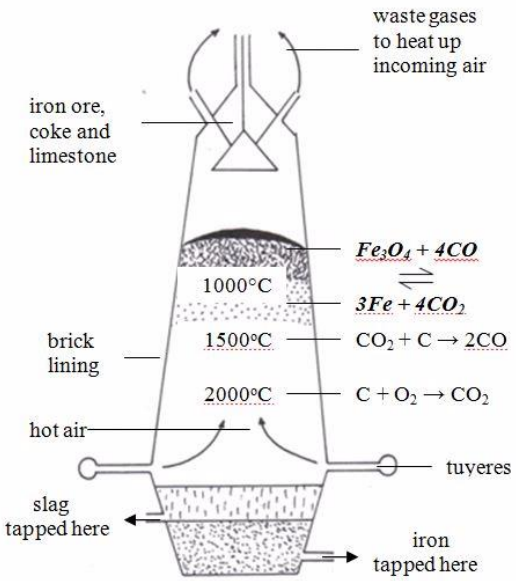
(iii) ∴Name of metal M is magnesium (Mg) *00½ Mark*

SECTION C (30 marks)

9.

(a) Aluminum is more reactive than iron, hence requires electrolytic method, while iron is moderately reactive and is extracted by reduction method. (02 marks)

(b) The blast furnace



$CaCO_{3(s)} \rightarrow CaO_{(s)} + CO_{2(g)}$
$CaO_{(s)} + SiO_{2(s)} \rightarrow CaSiO_{3(l)}$
Top: $Fe_2O_{3(s)} + 3CO_{(g)} \rightarrow 2Fe_{(s)} + 3CO_{2(g)}$
Middle: $C_{(s)} + CO_{2(g)} \rightarrow 2CO_{(g)}$
Bottom: $C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}$

Title: 01 mark
Diagram: 02 marks
Equations: 02 marks@

Total: 15 marks

10. (a) the doctor prescribes antacids tablets: this is to ensure neutralization of the excess stomach acid produced *02 Marks*

(b) $Mg(OH)_{2(aq)} + 2HCl_{(aq)} \rightarrow MgCl_{2(aq)} + 2H_2O_{(l)}$ *02 Marks*

(c) Excess stomach acid
210 mM – 160 mM = 50 mM *00½ Mark*
But:
1 molarity = 1000 milimolarity
? = 50 milimolarity

$$x = \frac{50\ milimolarity \times 1\ \text{milarity}}{1000\ milimolarity}$$

x = 0.05 M *00½ Mark* *00½ Mark*

Case i: find no. of moles of HCl

From

Molarity = $\frac{\text{no.of moles}}{vol.of\ solution}$ *00½ Mark*

No. of mole = 0.05 mol/dm^3 × 0.5 dm^3

= 0.025 moles *00½ Mark*

Case ii: find no. of moles of $Mg(OH)_2$

From, balanced chemical equation

$Mg(OH)_2 + 2HCl_{(aq)} \rightarrow MgCl_{2(aq)} + 2H_2O_{(l)}$

1 mole of $Mg(OH)_2 \equiv$ 2 moles of HCl *00½ Mark*

? $\equiv$ 0.025 moles of HCl

$= \frac{0.025\ \text{of HCl} \times 1\ \text{mol of Mg(OH)}_2}{2\ \text{moles of HCl}}$ *00½ Mark*

0.0125 mol of $Mg(OH)_2$ *00½ Mark*

Case iii: Find the mass of tablet

From

Mass of $Mg(OH)_2$ = no. of moles of $Mg(OH)_2$ × Mr of $Mg(OH)_2$ *00½ Mark*

= 0.0125 moles × 58g/mole *00½ Mark*

Mass of $Mg(OH)_2$ = 0.725g

∴the mass of $Mg(OH)_2$ 0.725g *00½ Mark*

(d) Find the number of tablets

From

1 tab = 145 mg of $Mg(OH)_2$

1g = 1000 mg

0.725g = x *00½ Mark*

x = 725 mg

1 tab $x = \frac{0.725g \times 1000mg}{1g}$ *00½ Mark*

= 145 mg of $Mg(OH)_2$

x = 725 mg of $Mg(OH)_2$ *00½ Mark*

$x = \frac{1\ \text{tab} \times 725mg\ of\ \text{Mg(OH)}_2}{145\ mg\ of\ \text{Mg(OH)}_2}$ *00½ Mark*

= 5 tablets

∴ the students should be given 5 tablets of $Mg(OH)_2$ *00½ Mark*

(e) Number of days to finish full dose

1 tablet = 12 hours

5 tablets = ?

$= \frac{5\ \text{tab} \times 12\ \text{hours}}{1\ \text{tab}}$ *00½ Mark*

= 60 hours *00½ Mark*

1 day = 24 hours

? = 60 hours *00½ Mark*

$= \frac{60\ \text{hours} \times 1\ \text{day}}{24\ \text{hours}} = 2.5$ days

∴ the student requires 2.5 days to finish the dose *00½ Mark*

Total: 15 marks

11. Characteristics of a good fuel

Introduction: Fuel is … (01½ marks)

Characteristics of a good fuel:

- It should be environmentally friendly (not harm the environment) in the course of its production and use, that is, it should not produce harmful or toxic products such as much smoke, carbon dioxide, carbon monoxide, sulphur dioxides, etc, which pollutes the air.
- It must be affordable to most people i.e. it must be cheap.
- It should not emit or produce dangerous by-products such as poisonous fumes, vapour or gases.
- It should have high calorific value i.e. it must burn easily and produce a tremendous quantity of heat energy per unit mass of the fuel.
- It should be easy and safe to transport, store, handle and use.
- It should be readily available in large quantities and easily accessible.
- It should have high pyrometric burning effect (highest temperature that can be reached by a burning fuel). Normally gaseous fuels have the highest pyrometric effect as compared to liquid and solid fuels.
- It should have a moderate velocity of combustion (the rate at which it burns) to ensure a steady and continuous supply of heat.
- A good fuel should have an average ignition point (temperature to which the fuel must be heated before it starts burning). A low ignition point is not good because it makes the fuel catch fire easily, which is hazardous, while high ignition point makes it difficult to start a fire with the fuel.
- A good fuel should have a low content of non-combustible material, which is left as ash or soot when the fuel burns. A high content of no-combustible material tends to lower the heat value of the fuel.

(Any eight @01½ = 12 marks)

Conclusion: There is no fuel among the different fuels known that posses all the virtues that a good fuel should have… (01½ marks)

(Total 15 marks)